
IC 242: Continuum Mechanics

Course Code: IC 242

Course Name: Continuum Mechanics

L-T-P-C: 2.5-0.5-0-3

Prerequisites: Consent of the faculty member

Students intended for: B.Tech

Elective or Core: Core

Approval: 2nd Senate

Course Contents

- **Introduction**

The Continuum Concept. [1 Lectures]

- **Tensor Analysis**

Tensor analysis in Cartesian coordinate, Gradient and Divergence, Dyadic and Dyadic algebra, Isotropic Tensor [4 Lectures]

- **Stress principles**

Cauchy stress, Principal stresses and principal direction of stress, Deviatoric stresses and their directions. [10 Lectures]

- **Fluid Statics**

Pascal's law, hydrostatic pressure, pressure measurement, manometer and micro-manometer, pressure gauge. [3 Lectures]

- **Kinematics**

Lagrangian and Eulerian description, Deformation gradient, deformation tensors, strain tensors, velocity gradient, rate of deformation. [4 Lectures]

- **Conservation laws**

Conservation of mass, conservation of linear momentum, moment of momentum, conservation of energy, Integral & differential approach and application to the control volume. Clausius- Duhem equality. [8 Lectures]

- **Constitutive theories**

Governing equations of a Continuum: Constitutive equations in material description, Elastic materials, Viscous fluids, Thermodynamic considerations [8 Lectures]

- **Elasticity**

linear elasticity and hyperelasticity (compressible and incompressible materials) [2 Lectures]

- **Plasticity**

Yield criteria, linear plasticity [2 Lectures]



References:

1. D. Frederick and T.S. Chang, **Continuum Mechanics** .
2. Philip G. Hodge, JR, **Continuum Mechanics** , Mc. Graw- Book Co.
3. A. C. Eringen, **Mechanics of Continuuua**, John Wiley & Sons.
4. chang, **Continuum Mechanics**, Prentice Hall, 1983.
5. Thomas, **Continuum Mechanics for Engineers**, CRC Press, 1999.
6. T. Mase, G. Mase, **Continuum Mechanics for Engineers**, CRC Press, 1999.
7. RM Bowen, **Introduction to Continuum Mechanics for Engineers**, Plenum Press, 1989.